
Practical 2: Pixel relationships (4-connected, 8-connected).

Aim: Pixel relationships (4-connected, 8-connected).

Software Required: GoogleColab

Code:

```
import numpy as np
import matplotlib.pyplot as plt
image = np.array([
    [0,0,1,0,0],
    [0,1,1,1,0],
    [1,1,1,1,1],
    [0,1,1,1,0],
    [0,0,1,0,0],
])
x, y = 2, 2
four_connected = [
    (x-y,y),
    (x+1,y),
    (x,y-1),
    (x,y+1),
]
eight_connected = [
    (x-1,y-1),
    (x-1,y),
    (x-1,y+1),
    (x,y-1),
    (x,y+1),
    (x+1,y-1),
```

```
(x+1,y),
(x+1,y+1),
]
four_connected = [
    p for p in four_connected
    if 0<=p[0]<image.shape[0] and 0<=p[1]<image.shape[1]
]
eight_connected = [
    p for p in eight_connected
    if 0<=p[0]<image.shape[0] and 0<=p[1]<image.shape[1]
]
print("selected pixel: ",(x,y))
print("\n4-connected neighbours:")
for p in four_connected:
    print(p,"value=",image[p])
print("\n8-connected neighbours:")
for p in eight_connected:
    print(p,"value=",image[p])
plt.imshow(image,cmap='gray')
plt.title("4-connected and 8-connected pixels")
plt.xticks(range(image.shape[1]))
plt.yticks(range(image.shape[0]))
plt.grid(True)
plt.scatter(y,x,s=150,facecolors='none',edgecolors='red')
plt.show()
```

Output:

selected pixel: (2, 2)

4-connected neighbours:

(0, 2) value= 1

(3, 2) value= 1

(2, 1) value= 1

(2, 3) value= 1

8-connected neighbours:

(1, 1) value= 1

(1, 2) value= 1

(1, 3) value= 1

(2, 1) value= 1

(2, 3) value= 1

(3, 1) value= 1

(3, 2) value= 1

(3, 3) value= 1

